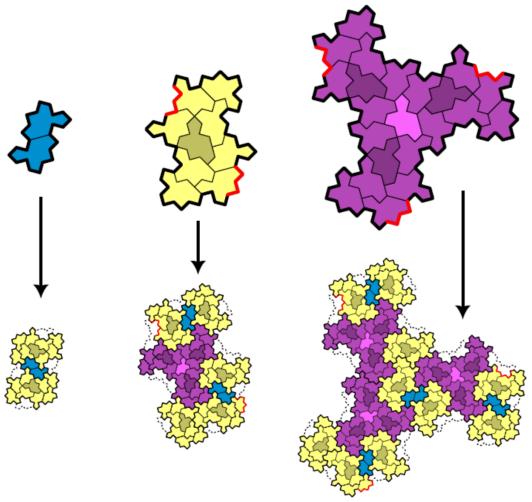
deflation for diabolo/2-arms/3-arms

Marianne Imperor-Clerc

CNRS, Université Paris Saclay
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numbers of diabolo/2-arms/3-arms metatiles using the 3x3 deflation matrix

The total number of each metatile at each iteration are calculated using a 3x3 deflation matrix 'mat' relating the numbers of metatiles at iteration i to that at iteration i+1.

The 3x3 deflation matrix

In[7]:=
```
mat = {{1, 2, 4}, {2, 5, 10}, {0, 1, 3}};
mat // MatrixForm
```

Out[8]//MatrixForm=

$$\begin{pmatrix} 1 & 2 & 4 \\ 2 & 5 & 10 \\ 0 & 1 & 3 \end{pmatrix}$$

Eigen values of 'mat'

In[9]:=
```
Eigenvalues[mat]
```

Out[9]=
$$\{4 + \sqrt{15}, 1, 4 - \sqrt{15}\}$$

Perron root is the maximum eigenvalue.
Its irrationality proves the quasiperiodicity.
Here it is a quadratic irrational number (periodic tail of the continuous fraction).

In[10]:=
```
perron = Eigenvalues[mat][[1]]
perron // N
ContinuedFraction[perron, 20]
```

Out[10]=
$$4 + \sqrt{15}$$

Out[11]=
```
7.87298
```

Out[12]=
```
{7, 1, 6, 1, 6, 1, 6, 1, 6, 1, 6, 1, 6, 1, 6, 1, 6, 1, 6, 1}
```

Eigen vectors of ‘mat’

In[13]:=
```
Eigenvectors[mat]
```

Out[13]=
$$\{\{2, 1 + \sqrt{15}, 1\}, \{-1, -2, 1\}, \{2, 1 - \sqrt{15}, 1\}\}$$

Eigen vectors associated to the Perron root giving the ratio of the three metatiles in the infinite tiling:
Diabolo/2-arms/3-arms == 2/1+\sqrt{15}/1 == 2/4.873/1

In[14]:=
```
Eigenvectors[mat][[1]]
```

Out[14]=
$$\{2, 1 + \sqrt{15}, 1\}$$

In[15]:=
```
Eigenvectors[mat][[1]] // N
```

Out[15]=
```
{2., 4.87298, 1.}
```

Choice of the initial numbers of metatiles for iteration: one diabolo D

In[16]:=
```
nD0 = 1;
nA20 = 0;
nA30 = 0;
v0 = {nD0, nA20, nA30};
```

In[42]:=
```
vtest = MatrixPower[mat, 1].v0
Print[vtest[[1]]]
Print[vtest[[2]]]
Print[vtest[[3]]]
```

Out[42]=
```
{4, 10, 3}
```

```
4
10
3
```

Numbers of metatiles for different inflation steps

In[46]:=
```
nmax = 10;

Do[v = MatrixPower[mat, n].v0;
 Print["{N_D, N_A2, N_A3} = ", v, "  N_A3/N_D = ", v[[1]] / v[[3]] // N]
 , {n, 1, nmax}]
```

```
{N_D, N_A2, N_A3} = {4, 10, 3}  N_A3/N_D = 1.33333
{N_D, N_A2, N_A3} = {36, 88, 19}  N_A3/N_D = 1.89474
{N_D, N_A2, N_A3} = {288, 702, 145}  N_A3/N_D = 1.98621
{N_D, N_A2, N_A3} = {2272, 5536, 1137}  N_A3/N_D = 1.99824
{N_D, N_A2, N_A3} = {17 892, 43 594, 8947}  N_A3/N_D = 1.99978
{N_D, N_A2, N_A3} = {140 868, 343 224, 70 435}  N_A3/N_D = 1.99997
{N_D, N_A2, N_A3} = {1 109 056, 2 702 206, 554 529}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {8 731 584, 21 274 432, 4 365 793}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {68 743 620, 167 493 258, 34 371 811}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {541 217 380, 1 318 671 640, 270 608 691}  N_A3/N_D = 2.
```

Choice of the initial numbers of metatiles for iteration: one 2-arms A2

In[23]:=
```
nD0 = 0;
nA20 = 1;
nA30 = 0;
v0 = {nD0, nA20, nA30};
```

In[27]:=
```
MatrixPower[mat, 1].v0
```

Out[27]=
```
{2, 5, 1}
```

Numbers of metatiles for different inflation steps

In[48]:=
```
nmax = 10;

Do[v = MatrixPower[mat, n].v0;
 Print["{N_D, N_A2, N_A3} = ", v, "  N_A3/N_D = ", v[[1]] / v[[3]] // N]
 , {n, 1, nmax}]
```

```
{N_D, N_A2, N_A3} = {4, 10, 3}  N_A3/N_D = 1.33333
{N_D, N_A2, N_A3} = {36, 88, 19}  N_A3/N_D = 1.89474
{N_D, N_A2, N_A3} = {288, 702, 145}  N_A3/N_D = 1.98621
{N_D, N_A2, N_A3} = {2272, 5536, 1137}  N_A3/N_D = 1.99824
{N_D, N_A2, N_A3} = {17 892, 43 594, 8947}  N_A3/N_D = 1.99978
{N_D, N_A2, N_A3} = {140 868, 343 224, 70 435}  N_A3/N_D = 1.99997
{N_D, N_A2, N_A3} = {1 109 056, 2 702 206, 554 529}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {8 731 584, 21 274 432, 4 365 793}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {68 743 620, 167 493 258, 34 371 811}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {541 217 380, 1 318 671 640, 270 608 691}  N_A3/N_D = 2.
```

Choice of the initial numbers of metatiles for iteration: one 3-arms A3

In[30]:=
```
nD0 = 0;
nA20 = 0;
nA30 = 1;
v0 = {nD0, nA20, nA30};
```

In[34]:=
```
MatrixPower[mat, 1].v0
```

Out[34]=
```
{4, 10, 3}
```

Numbers of metatiles for different inflation steps

In[50]:=
```
nmax = 10;

Do[v = MatrixPower[mat, n].v0;
 Print["{N_D, N_A2, N_A3} = ", v, "  N_A3/N_D = ", v[[1]] / v[[3]] // N]
 , {n, 1, nmax}]
```

```
{N_D, N_A2, N_A3} = {4, 10, 3}  N_A3/N_D = 1.33333
{N_D, N_A2, N_A3} = {36, 88, 19}  N_A3/N_D = 1.89474
{N_D, N_A2, N_A3} = {288, 702, 145}  N_A3/N_D = 1.98621
{N_D, N_A2, N_A3} = {2272, 5536, 1137}  N_A3/N_D = 1.99824
{N_D, N_A2, N_A3} = {17 892, 43 594, 8947}  N_A3/N_D = 1.99978
{N_D, N_A2, N_A3} = {140 868, 343 224, 70 435}  N_A3/N_D = 1.99997
{N_D, N_A2, N_A3} = {1 109 056, 2 702 206, 554 529}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {8 731 584, 21 274 432, 4 365 793}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {68 743 620, 167 493 258, 34 371 811}  N_A3/N_D = 2.
{N_D, N_A2, N_A3} = {541 217 380, 1 318 671 640, 270 608 691}  N_A3/N_D = 2.
```